

KEY

Statics Practice Exam 4
(Includes problems from Fall 2022 plus other semesters)

Section 1 Fluid Pressure

1. (5 pts) The concrete dam is designed so that its face AB has a gradual slope into the water as shown. The dam is 50 ft wide (into the page) and has a height h of 8 ft. Water has a specific weight of 62.4 lb/ft^3 . What is the magnitude of the horizontal force component the water exerts on face AB of the dam?

- A. 100 kip ✓
- B. 90 kip
- C. 96 kip
- D. 87 kip
- E. 103 kip

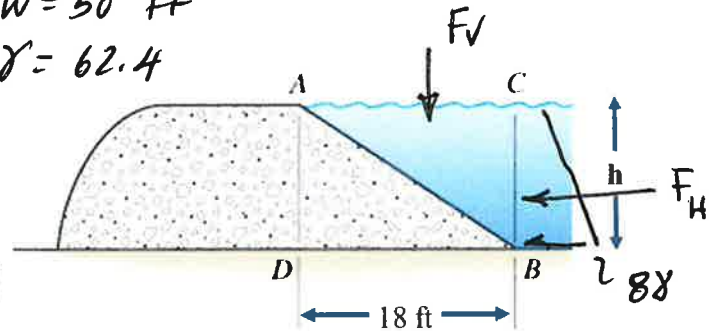
$$F_H = \frac{1}{2} b h w$$

$$= \frac{1}{2} (8)(62.4)(50)$$

$$F_H = 99.8 \text{ kip}$$

$$W = 50 \text{ ft}$$

$$\gamma = 62.4$$



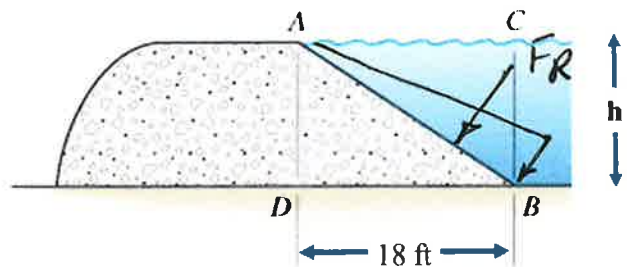
2. (5 pts) The concrete dam is designed so that its face AB has a gradual slope into the water as shown. The dam is 60 ft wide (into the page) and has a height h of 9 ft. Water has a specific weight of 62.4 lb/ft^3 . Where would the center of pressure be located for the resultant force **measured from A along face AB**?

- A. 13.4 ft ✓
- B. 12.9 ft
- C. 16.8 ft
- D. 15.2 ft
- E. 14.4 ft

$$d = \sqrt{9^2 + 18^2}$$

$$= 20.1 \text{ ft}$$

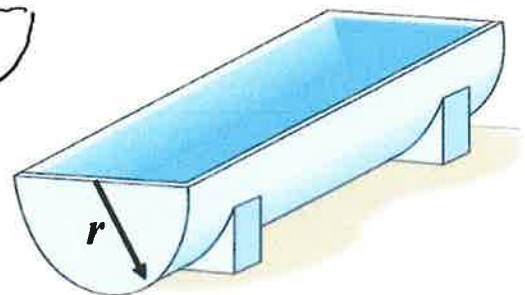
$$\text{From A: } \frac{2}{3}(20.1) = 13.42 \text{ ft}$$



3. (5 pts) The storage tank is filled to the top with a liquid that has a density of 900 kg/m^3 . Determine the resultant force that it exerts on the semi-circular end plate if $r = 0.8 \text{ m}$.

- A. 3.01 kN ✓
- B. 2.83 kN
- C. 2.02 kN
- D. 3.29 kN
- E. 4.04 kN

$$\frac{4R}{3\pi} = \bar{z}$$



$$F_R = (P \text{ at } \bar{z}) * \text{Area}$$

$$= \gamma \cdot \bar{z} \cdot A$$

$$= (900)(9.81) \left(\frac{4(0.8)}{3\pi} \right) \left(\frac{\pi}{2} (0.8)^2 \right) = 3.01 \text{ kN}$$

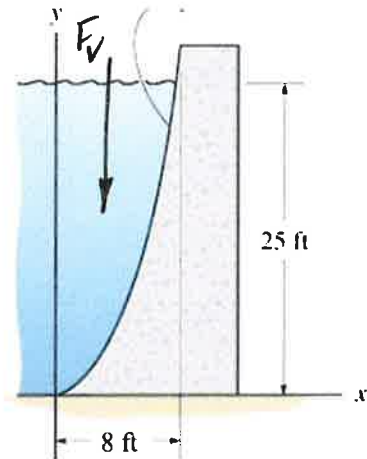
4. (5 pts) What is the total vertical component of force the water exerts on the side of the parabolic shaped dam? The dam is 15 ft wide (into the page or screen) and the specific weight of water is 62.4 lb/ft³.

- A. 125 kip
- B. 205 kip
- C. 160 kip
- D. 178 kip
- E. 112 kip

$$F_v = W =$$

$$= \frac{2}{3}(8)(25)(15)(62.4)$$

$$F_v = 124.8 \text{ kip}$$



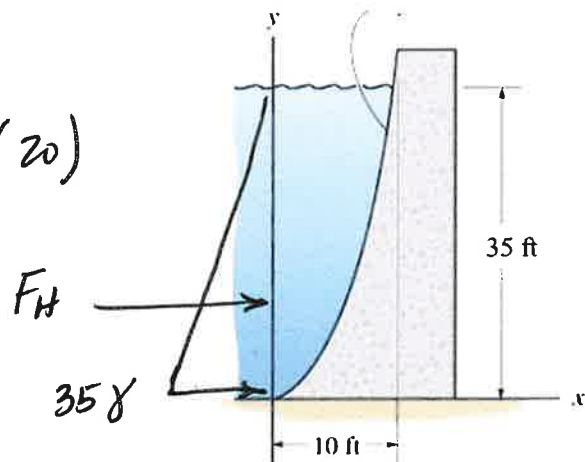
5. (5 pts) What is the total horizontal component of force the water exerts on the side of the parabolic shaped dam? The dam is 20 ft wide (into the page or screen) and the specific weight of water is 62.4 lb/ft³.

- A. 764 kip
- B. 805 kip
- C. 660 kip
- D. 878 kip
- E. 712 kip

$$F_H = \frac{1}{2} b h w$$

$$= \frac{1}{2}(35)(35)(20)$$

$$F_H = 764.4 \text{ kip}$$



MOI Problems

6. Determine I_y for the area shown at right.

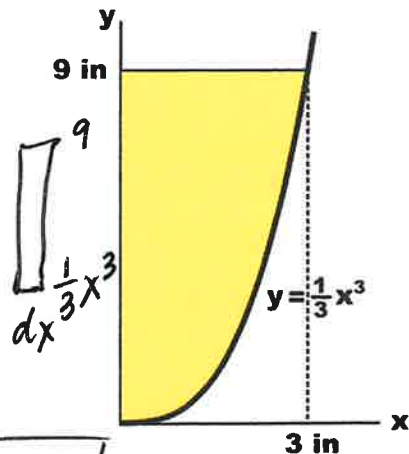
- A. 40.5 in⁴
- B. 30.5 in⁴
- C. 20.5 in⁴
- D. 10.5 in⁴
- E. 27.0 in⁴

$$I_y = \int_A \tilde{x}^2 dA$$

$$= \int_0^3 x^2 \left(9 - \frac{1}{3}x^3\right) dx$$

$$= \int_0^3 \left(9x^2 - \frac{1}{3}x^5\right) dx$$

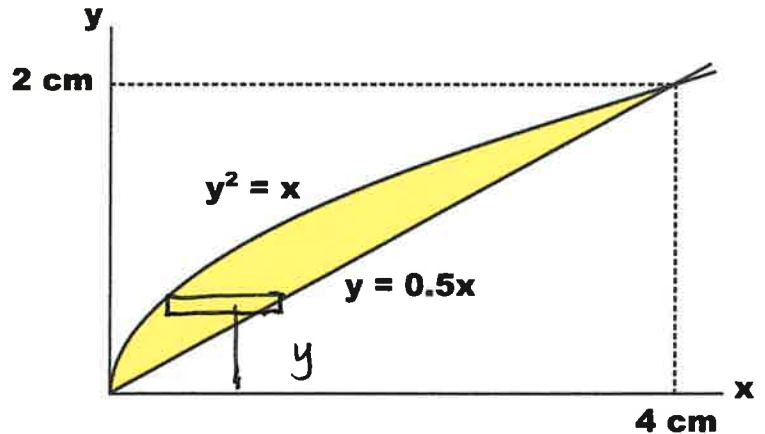
$$= \left(\frac{9}{3}x^3 - \frac{1}{18}x^6\right) \Big|_0^3 = 40.5 \text{ in}^4$$



TI-89 $\int (x^2 * (9 - .333 * x^3), x, 0, 3) = 40.54$

7. For the area shown below defined by the two curves, to find I_x by integration you use a dA element parallel to the x axis. The expression in the integral is the following:

- A. $[y^2(2y - y^2)]dy$
- B. $[y^2(y^2 - y)]dy$
- C. $[x^2(y^2 - y)]dy$
- D. $[x^2(2y - y^2)]dx$
- E. $[y^2(10y^{2.5} - y^{0.5})]dy$



$$= \int y^2 x_R - x_L dy = \int y^2 \left(\frac{y}{0.5} - y \right) dy = \int y^2 (2y - y) dy = \int y^3 dy$$

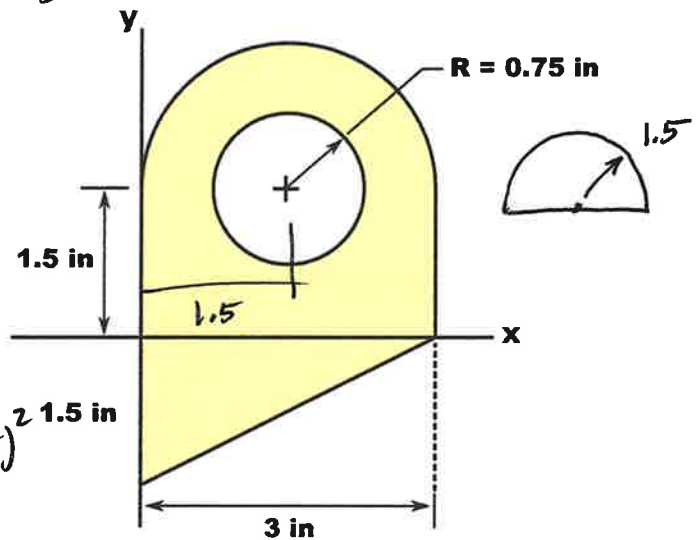
This composite shape is composed of a rectangle, a half-circle, a triangle and a circular hole.

8. Determine I_y for the half-circle.

- A. 9.94 in⁴
- B. 5.94 in⁴
- C. 15.94 in⁴
- D. 12.94 in⁴
- E. 7.94 in⁴

$$I_y = \bar{I} + Ad^2 = \frac{1}{8}\pi(1.5)^4 + \frac{\pi}{2}(1.5)^2(1.5)^2$$

$$I_y = 9.94 \text{ in}^4$$



This composite shape is composed of a rectangle, a half-circle, a triangle and a circular hole.

9. Determine I_x for the triangle.

- A. 0.844 in⁴
- B. 4.5 in⁴
- C. 2.25 in⁴
- D. 0.281 in⁴
- E. 1.38 in⁴

$$I_x = I_{\text{Base}} = \frac{1}{12}(3)(1.5)^3$$

$$I_x = 0.844 \text{ in}^4$$

This composite shape is composed of a rectangle, a half-circle, a triangle and a circular hole.

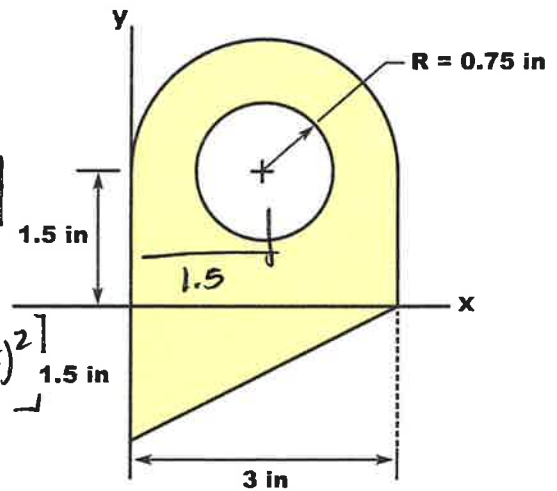
10. Determine I_y for the hole.

- A. -4.22 in⁴
- B. -3.96 in⁴
- C. -0.249 in⁴
- D. -2.25 in⁴
- E. -1.72 in⁴

$$I_y = - [\bar{I} + A d^2]$$

$$= - \left[\frac{\pi}{4} (.75)^4 + \pi (.75)^2 (1.5)^2 \right]$$

$$I_y = -4.225 \text{ in}^4$$



Section 3 – Product of Inertia

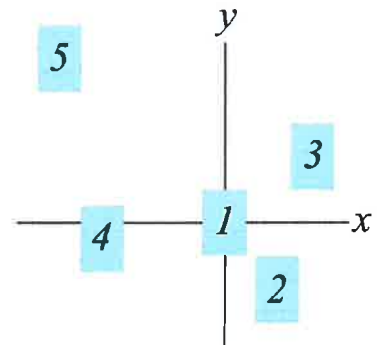
11. (5 pt) Determine the following statement that is false.

- ✓ a. Since the product of inertia equation uses both centroidal dimensions, its sign is always positive.
- b. Products of inertia have the same units as moments of inertia.
- c. The farther an area lies away from either axis, the larger the product of inertia magnitude will be.
- d. The closer an area lies to either axis, the smaller the product of inertia magnitude will be.
- e. A product of inertia cannot be calculated with respect to only a single axis.

I_{xy}
can be
+ or -

12. (5 pt) If the rectangles are of equal area and the centroid of area 1 is at the origin, which area(s) have non-zero product(s) of inertia?

- ✓ a. 2, 3, 4, 5
- b. 1
- c. 3, 4
- d. 2, 5
- e. 1, 2, 3
- f. 1, 3, 5
- g. 1, 2, 3, 4, 5



13. (5 pt) Determine the integrand for the product of inertia calculation about the x and y axes for the shaded area, integrating with respect to x .

- a. $x^2 + x/2$
- b. $(1/3)(2x+1)^{3/2}$
- c. $(1/12)(2x+1)^{3/2}$
- d. $2x + 1/2$
- e. $(2x+1)^{1/2}$

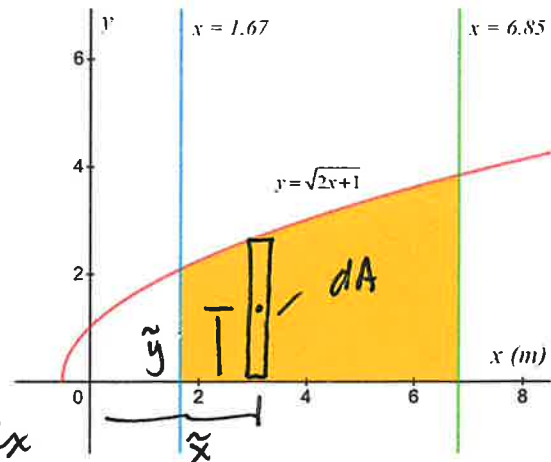
Tricky!

$$I_{xy} = \int_A \tilde{x} \tilde{y} dA$$

$$= \int_A x \left(\frac{1}{2} \right) (\sqrt{2x+1}) (\sqrt{2x+1}) dx$$

$$= \int_A \frac{x}{2} (2x+1) dx$$

$$= \int_A (x^2 + x/2) dx$$



$$dA = (\sqrt{2x+1} - 0) dx$$

$$\tilde{x} = x$$

$$\tilde{y} = \frac{1}{2} y = \frac{1}{2} \sqrt{2x+1}$$

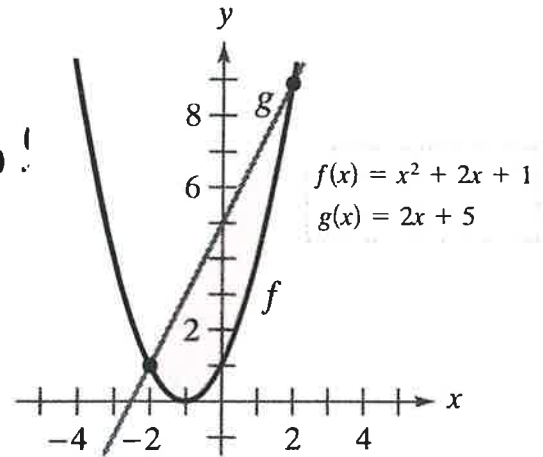
14. (5 pt) Determine the product of inertia with respect to the x and y axes for the area bound between the two functions shown in the region $-2 \leq x \leq 2$ ft.

- a. 17.07 ft⁴
- b. 16.84 ft⁴
- c. 16.16 ft⁴
- d. 17.52 ft⁴
- e. 18.20 ft⁴

I don't like this!

$$I_{xy} = \int_A \tilde{x} \tilde{y} dA$$

$$= \int_{-2}^2 \int_{x^2+2x+1}^{2x+5} xy \, dy \, dx$$



TI-89: $\int(\int(x*y, y, x^2 + 2*x + 1, 2*x + 5), x, -2, 2)$
 $= 17.07 \text{ ft}^4$

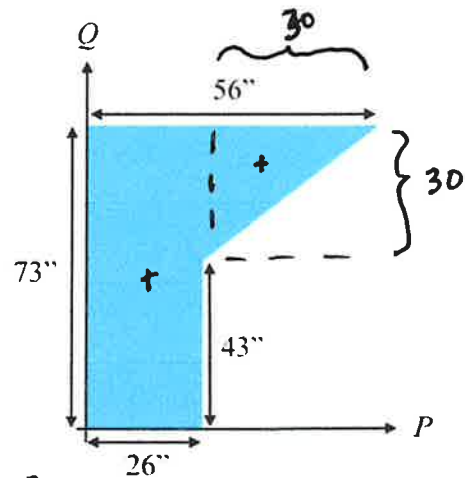
15. (5 pt) Determine the product of inertia for the shaded area shown about the P-Q axes.

- a. 1.932 (10⁶) in⁴
- b. 2.528 (10⁶) in⁴
- c. 2.302 (10⁶) in⁴
- d. 1.819 (10⁶) in⁴
- e. 2.224 (10⁶) in⁴

Touching origin

$$I_{xy} = \frac{1}{4} b^2 h^2$$

$$\bar{I} = \frac{1}{72} b^2 h^2$$



Section 4 – Mohr's Circle Transformations

16. (5 pt) Determine the following statement that is false regarding Mohr's Circle.

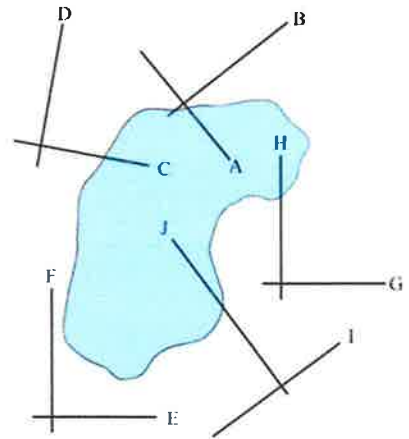
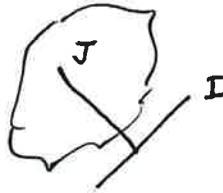
- a. For statics applications, there are reference points located on the circumference of Mohr's Circle that will lie to the left of the product of inertia axis.
- b. An angle of 120 degrees between two reference points on Mohr's Circle equates to 60 degrees between their corresponding axes on the cross-section.
- c. Principal moments of inertia and non-principal moments of inertia can both be calculated with Mohr's Circle, given the appropriate reference information.
- d. The average moment of inertia between two perpendicular reference axes will always locate the center of Mohr's Circle for that area and origin.
- e. Horizontal dimensions measured from the origin on Mohr's Circle represent moment of inertia values on the cross-section.

$$I_{xy} = \frac{1}{4} (26)^2 (73)^2 + \left[\underbrace{\frac{1}{72} (30)^2 (30)^2}_{\bar{I}} + \underbrace{\frac{1}{2} (30) (30)}_A \underbrace{(36)}_{dx} \underbrace{(63)}_{dy} \right]$$

$$I_{xy} = 1.9325 \cdot 10^6 \text{ in}^4$$

17. (5 pt) If only one of the following axes systems is principal for its respective origin, which system shows principal axes?

- ✓ f. I-J
g. A-B
h. C-D
i. E-F
j. G-H



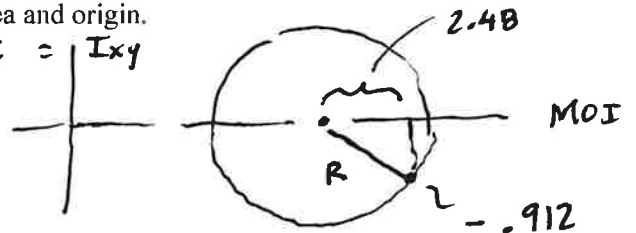
J is the only axis that passes near the center of the area, thus it must be I_{min} .

Then I_{max} is $\perp$ to I_{min} .

18. (5 pt) If the moment of inertia about a particular axis is $2.48(10^6) \text{ mm}^4$ larger than the center of Mohr's Circle and the product of inertia associated with that axis and another perpendicular axis is $-0.912(10^6) \text{ mm}^4$, determine the radius of Mohr's Circle for that area and origin.

- k. $2.64(10^6) \text{ mm}^4$
l. $2.48(10^6) \text{ mm}^4$
m. $1.24(10^6) \text{ mm}^4$
n. $1.69(10^6) \text{ mm}^4$
o. 0 mm^4

$POI = I_{xy}$

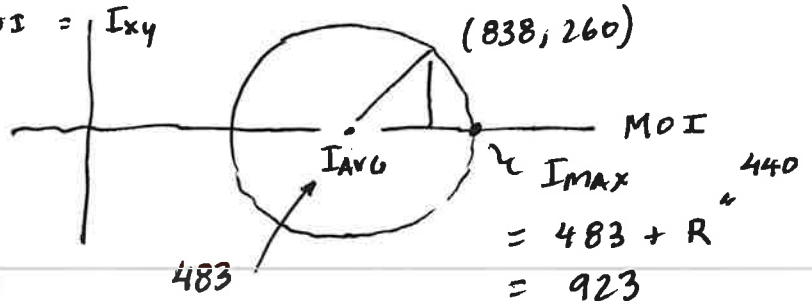


$$R = \sqrt{2.48^2 + .912^2} = 2.64 \cdot 10^6$$

19. (5 pt) If the average moment of inertia between two perpendicular reference axes for a given area is 483 in^4 and a reference point on Mohr's Circle lies at $(838, 260) \text{ in}^4$, determine the largest principal moment for this same area and origin.

- ✓ p. 923 in^4
q. 949 in^4
r. 896 in^4
s. 830 in^4
t. 795 in^4

$POI = I_{xy}$



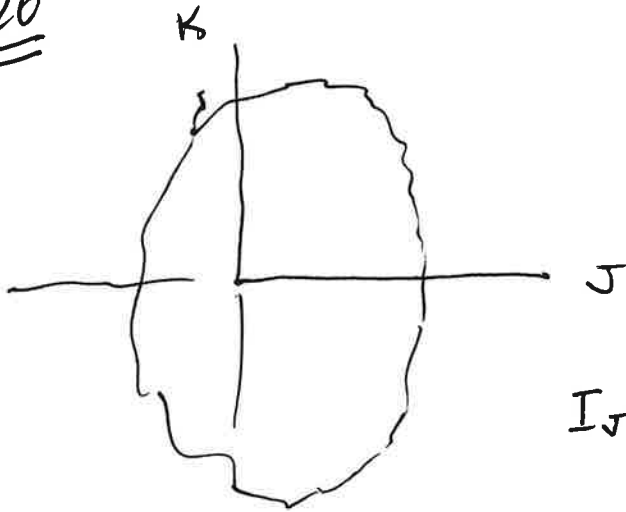
$$R = \sqrt{355^2 + 260^2} = 440$$

20. (5 pt) An area is symmetric about axis J but not K (J-K lie perpendicular to each other), and the moment of inertia about axis J is 371 in^4 while the moment of inertia about axis K is 60 in^4 . Determine the magnitude of the product of inertia about a set of axes rotated 62 degrees clockwise on Mohr's Circle from the reference points for J-K.

- u. 137.3 in^4
v. 149.6 in^4
w. 154.0 in^4
x. 120.1 in^4
y. 113.8 in^4

See next page.

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$$I_K = 60 \text{ in}^4$$

$$I_J = 371 \text{ in}^4$$

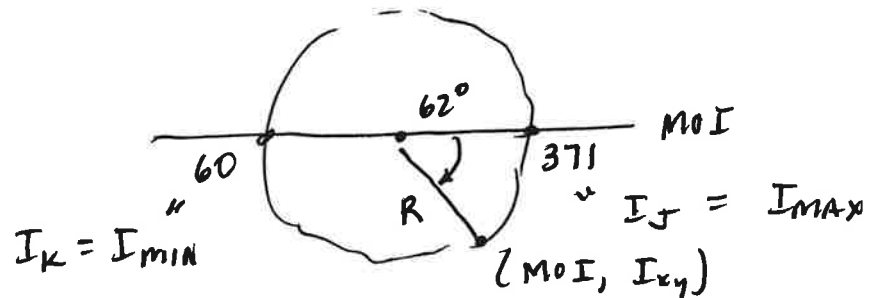
Symmetric about J, 0°

$$I_{xy} = 0$$

and

$$I_J = I_{\max}$$

$$I_K = I_{\min}$$



$$R = \frac{371 - 60}{2} = 155.5$$

$$\begin{aligned} I_{xy} &= -155.5 \cdot \sin 62 \\ &= -137.3 \text{ in}^4 \end{aligned}$$